

Helium (He)

A colorless, odorless, nonflammable, inert gas.

GRADES & SPECIFICATIONS

Grade	% Purity	O ₂	H ₂ O	THC	CO	CO ₂	Ar	N ₂ (*)
Research Ultra 6.0	99.9999	10 ppb	20 ppb	100 ppb	—	—	—	5
Research 6.0 ⁽¹⁾	99.9999	0.5	0.2	0.1	0.1	0.1	0.5	0.5
Chromatographic 6.0	99.9999	0.5	0.2	0.1	0.1	0.1	0.5	0.5
Ultra-Pure Carrier (UPC)	99.9995	1	1	0.5	1 ⁽²⁾	1 ⁽²⁾	—	3
Semiconductor 5.0	99.999	1	1	0.5	1 ⁽²⁾	1 ⁽²⁾	—	5
Ultra High Purity (UHP) 5.0	99.999	1	1	0.5	1 ⁽²⁾	1 ⁽²⁾	—	5
High Purity (HP)	99.998	5	—	—	—	—	—	3

Concentrations given are ppm by volume unless otherwise specified.

Purity levels given excluding Neon. **(1)** Total of impurities listed shall not exceed 1 ppm. **(2)** Total CO + CO₂ shall not exceed 1 ppm. **(*)** Nitrogen spec for reference only.

Individual or Batch Certificate of Analysis available upon request.

ORDERING INFORMATION

Product	Cyl. Size	Contents (ft ³)	Valve (CGA)	Product Number	Pressure @70°F (psig)	Delivery Range (psig)
Research Ultra 6.0	300	291	580	HE-RU-300	2,640	2-stage · 2–75
Research 6.0	300	291	580	HE-RES-300	2,640	Single / 2-stage · 0–150
	200	213	580	HE-RES-200	2,200	
	80	84	580	HE-RES-80	2,200	
	35	33	580	HE-RES-35	2,200	
Chromatographic 6.0	300	291	580	HE-CH-300	2,640	Single / 2-stage
	200	213	580	HE-CH-200	2,200	
	80	84	580	HE-CH-80	2,200	
	35	33	580	HE-CH-35	2,200	
Ultra-Pure Carrier (UPC)	300	291	580	HE-UPC-300	2,640	Single-stage · 0–100
	200	213	580	HE-UPC-200	2,220	
	80	84	580	HE-UPC-80	2,200	
	35	33	580	HE-UPC-35	2,200	
Semiconductor 5.0	300	291	580	HE-SEMI-300	2,640	2-stage
	200	213	580	HE-SEMI-200	2,200	

Continued on page 2 → additional grades & Technical Data

Helium (He)

Continued — Ordering Information & Technical Data

— ORDERING INFORMATION (CONT.)

Product	Cyl. Size	Contents (ft ³)	Valve (CGA)	Product Number	Pressure @70°F (psig)	Delivery Range (psig)
Ultra High Purity (UHP) 5.0	300	291	580	HE-UHP-300	2,640	2-stage · 0–100
	200	213	580	HE-UHP-200	2,200	
	80	84	580	HE-UHP-80	2,200	
	35	33	580	HE-UHP-35	2,200	
High Purity (HP)	6K	512	677	HE-HP-6K	6,000	200–3,500
	3K	328	680	HE-HP-3K	3,500	200–1,500

— TECHNICAL DATA & SHIPPING INFORMATION

Molecular Weight	4.0026 g/mol
Specific Volume	96.65 cf/lb @ 70°F & 1 ATM
Boiling Point	–268.9 °C
Flammability in Air	Nonflammable

U.S. DOT Name	Helium, Compressed
ID Number	UN 1046
DOT Hazard Class	2.2 — Nonflammable Gas
CAS Registry Number	7440-59-7

WSM offers a wide range of gas supply modes, including high-pressure cylinders, liquid cylinders (dewars), MicroBulk and bulk. Your WSM specialty gas representative can help you select the option best suited to your operation, including automatic changeover systems for uninterrupted supply.

WSM Quality Policy. The purpose of the WSM Quality System is to continually improve our manufacturing and related processes to provide our customers with the highest product purity, consistency, and service.